

JOSEPH MAINA NDEGWA

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BACKGROUND

A motivated Information Technology student seeking with practical experience in backend development, cybersecurity, and network engineering. Skilled in building and managing real-world systems using Python, Django, and Linux, with hands-on exposure to self-hosted infrastructure, IoT-based data collection systems, and API development. Strong understanding of networking concepts including VLANs, OSPF, and IPv6. Passionate about designing secure and scalable systems, solving technical problems, and continuously applying engineering skills in real-world environments.

KEY COMPETENCIES

Backend Development	Network Engineering	Problem Solving & Analytical
Linux Server Administration	Database Management (SQL)	Thinking
API & Integration	Self Hosted Infrastructure	Continuous Learning &
System Design & Architecture	Version Control(Git, Github)	Technical Adaptability

TECHNICAL EXPERIENCE

Local home lab May 2026

Self-Hosted Infrastructure Lab

Lead design and implementation of self-hosted infrastructure in a home lab environment focused on production-grade system administration and networking concepts.

Accomplishments:

- Installed VMware virtual machine.
- Designed and deployed Ubuntu-based mail server using iRedMail.
- Configured DNS, SMTP,IMAP, POP3 and TLS security protocols.
- Implemented server hardening and authentication mechanisms.
- Built hands-on expertise in Linux server administration and email infrastructure.

Public Procurement Transparency Dashboard Sep 2024 - Dec 2024

Full-Stack Project(Personal Project)

Designed and built a full-stack web application for data transparency and visualization.

Accomplishments:

- Developed Django REST API backend and React frontend.
- Designed PostgreSQL database schema for structured data storage.
- Built analytics dashboard for data visualization and reporting
- Implemented scalable API architecture for future expansion.

Enterprise Network Design Lab Year 2025

Network Engineering Simulation (Packet Tracer)

Designed and simulated enterprise-grade network architecture for learning routing and segmentation concepts.

Accomplishments:

- Configured VLAN segmentation across multiple switches
- Implemented OSPFv3 dynamic routing
- Designed IPv6 addressing scheme for scalability
- Simulated enterprise network topology for real-world scenarios

EDUCATION

Bachelor of Science in Business Information Technology

Muranga University of Technology - Muranga, Kenya.

2023 - Present

- Relevant Coursework: Data Structures & Algorithms, Networking, Database Systems Management, Operating Systems, System analysis and Design, Software Development, Distributed Systems, Software Engineering, Web Application Development, Simulation and Modelling etc.
- Key Projects: Network Simulation Labs, Front-end and Back-end development, API development, Programming and Database practicums.
- Focus: Back-end Development, Cybersecurity, Network Engineering and Automation

CERTIFICATIONS

- Data Analysis Essentials - Cisco
- Operating Systems - Cisco NetaCad
- SPSS & Stata Data Analysis - University Program
- AWS AI Practitioner - Udacity
- Kali Linux Basics - Simplilearn (SkillUp)
- n&n: no Code AI Agent Builder: Simplilearn

LANGUAGES

- English
- Kiswahili
- Gikuyu

REFERENCES

Available on request.